Chapter 18 Notes

Section 2

Lemma 2.2

Here is why $\cup_{q_0 \in \mathcal{G}} S(q_0)$ is an M-set (and hence so is S). Let $G(x, q)$ stand for the formula

$$(\forall y \in x)(\text{Ordinal}(y) \wedge (\exists g \in R_\beta^{\mathcal{G}})(q \Vdash [\![\hat{y} \approx g]\!]))$$

(this can be expanded to a real first-order formula by results in the previous chapter). If $S(q_0)$ happens to be in M, then $G(S(q_0), q_0)$ will be true in M. Now, we already know that each $S(q_0) \in M$. Then consider the following function which is definable over M:

$$x \in \mathcal{G} \wedge G(y, x)$$

Because the domain, $\mathcal{G}$, is in M, so is the range, and because we have shown each $S(q_0)$ to be in M, the range is the set of all $S(q_0)$ (for $q_0 \in \mathcal{G}$). Then the union of this, which is also in M, is just S itself.

Theorem 2.3

It is not fully clear to me why $p \Vdash [\![(\text{Ordinal}(f) \wedge \text{Ordinal}(\hat{\delta})) \supset (f \in \hat{\delta} \vee \hat{\delta} \in f \vee f \approx \hat{\delta})]\!]$.

A familiar error (see my notes on the previous chapter for what this error is) occurs in this proof.

Section 3

Theorem 3.2

A familiar error (see my notes on the previous chapter for what this error is) occurs in this proof.

Section 4

Notes

It should be noted why both $\mathcal{G} \in M$ and $\mathcal{R} \in M$ (which are both conditions we need to have). $\mathcal{G}$ is a subset of $(\mathcal{P} \cap M) \times (\mathcal{P} \cap M)$, which is in M. Also, $\mathcal{G}$ is definable over M by

$$(\exists P)(\exists N)(\exists n)(\exists m)(\exists f)(\exists g)(x = \langle P, N \rangle \wedge \text{Num}(n) \wedge \text{Num}(m) \wedge$$

$$\text{f is 1-1 func}(n, N) \wedge \text{g is 1-1 func}(m, P) \wedge P \cap N = \emptyset)$$

(this works because, for each 1-1 function from a number n to some N or P, it is finite, so it is in M). Thus $\mathcal{G} \in M$. As for $\mathcal{R}$, it is a subset of $\mathcal{G} \times \mathcal{G}$, which is in M, and is defined over M by the formula:

$$(\exists P_1, P_2)(\exists N_1, N_2)(x = \langle P_1, N_1 \rangle \wedge y = \langle P_2, N_2 \rangle \wedge x \in \mathcal{G} \wedge y \in \mathcal{G}$$

$$\wedge P_1 \subseteq P_2 \wedge N_1 \subseteq N_2)$$

and thus $\mathcal{R} \in M$.

In Case 1 of Theorem 4.1, note that $r' \vDash [\![\sim(\hat{n} \in \hat{c})]\!]$ would follow by Proposition 3.3 of the previous chapter.

Exercise 4.1

Suppose $n \in N$. We show that $\langle P, N \rangle \vDash [\![\sim(\hat{n} \epsilon f_0)]\!]$. Because $n \in N$, $n \notin P$. Thus $\langle P, N \rangle \vDash \sim(\hat{n} \epsilon f_0)$. Now, if $\langle P, N \rangle \mathcal{R} \langle P_1, N_1 \rangle$, then $N \subseteq N_1$, and thus $n \in N_1$, but $n \notin P_1$, which would mean $\langle P_1, N_1 \rangle \vDash \sim(\hat{n} \epsilon f_0)$. This explains why $\langle P, N \rangle \vDash \Box\sim(\hat{n} \epsilon f_0)$, and thus $\langle P, N \rangle \vDash [\![\sim(\hat{n} \epsilon f_0)]\!]$.